

Fixed width definition

Definition For b an integer greater than 1, w a positive integer, and n a nonnegative integer _____, the **base b fixed-width w expansion of n** is

$$(a_{w-1} \cdots a_1 a_0)_{b,w}$$

where $a_0, a_1, \dots, a_{w-1}$ are nonnegative integers less than b and

$$n = \sum_{i=0}^{w-1} a_i b^i$$

Representing zero

Representing 0: Which of the representations so far allow us to represent 0?